

Network Flow Problems

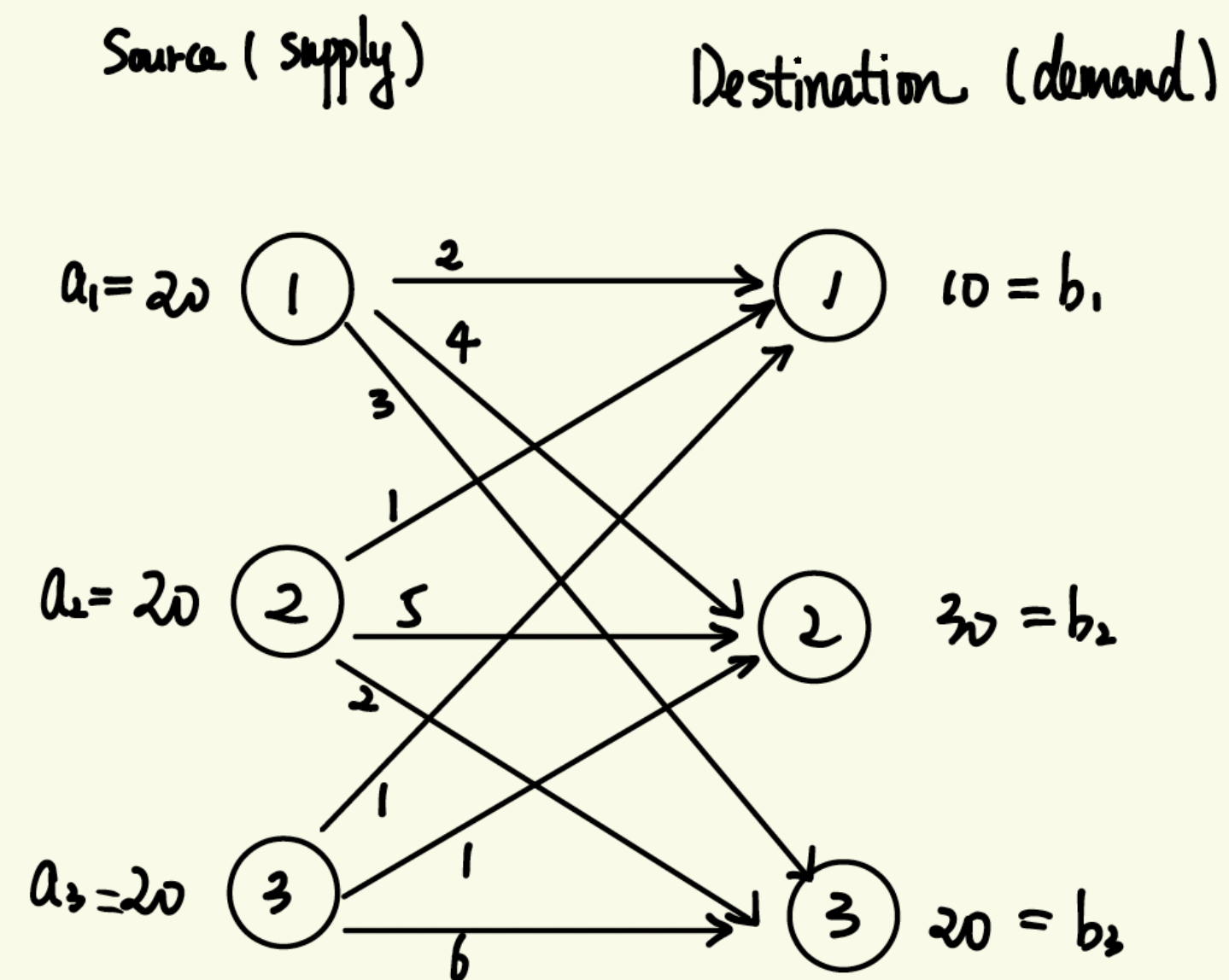
DSA 8420

Boshi Yang

Review

Transportation problem

- Ships identical goods from sources to destinations with unit shipping costs c_{ij} .
- Goal: Design a shipping program that matches supplies and demands with minimum total transportation cost.



a_i = # units available at Source i .

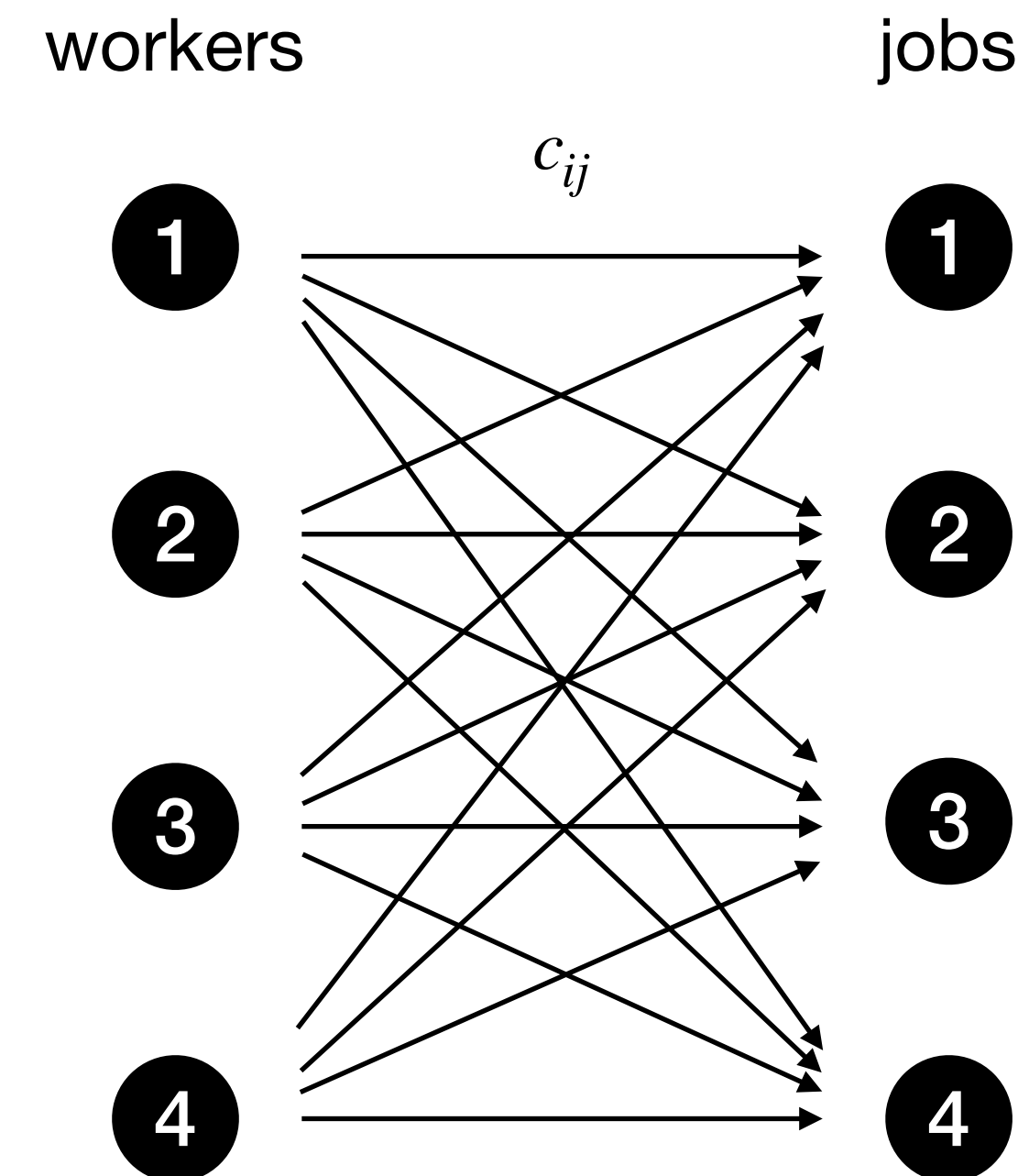
b_j = # units required at Destination j .

c_{ij} = cost of shipping one unit from source i to destination j .

Review

Assignment problem

- Assign workers to jobs. Each assignment incurs cost c_{ij} .
- Goal: Design one-to-one assignments that match workers and jobs with minimum total cost.

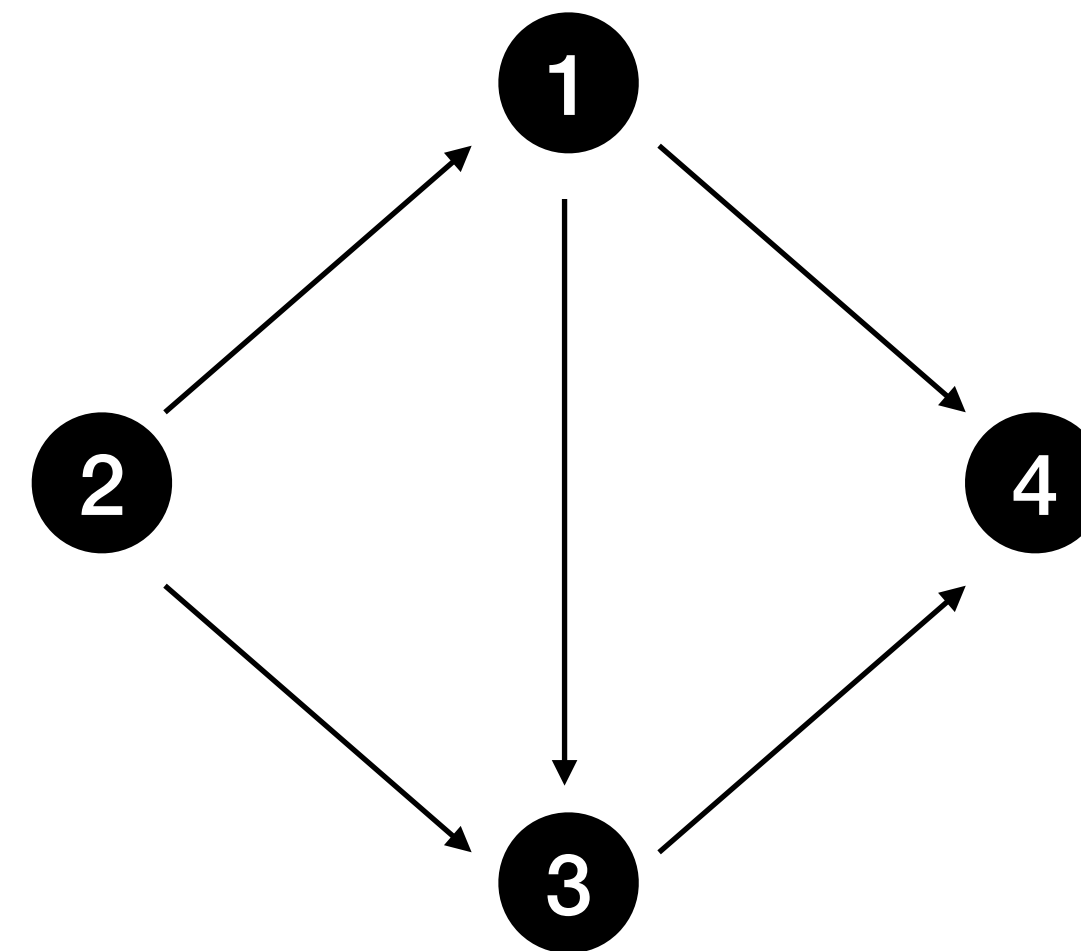


Directed Graph

Nodes and arcs

A directed graph $G = (\mathcal{N}, \mathcal{A})$

- $\mathcal{N}$: set of nodes: i
- $\mathcal{A}$: set of directed arcs: (i, j)



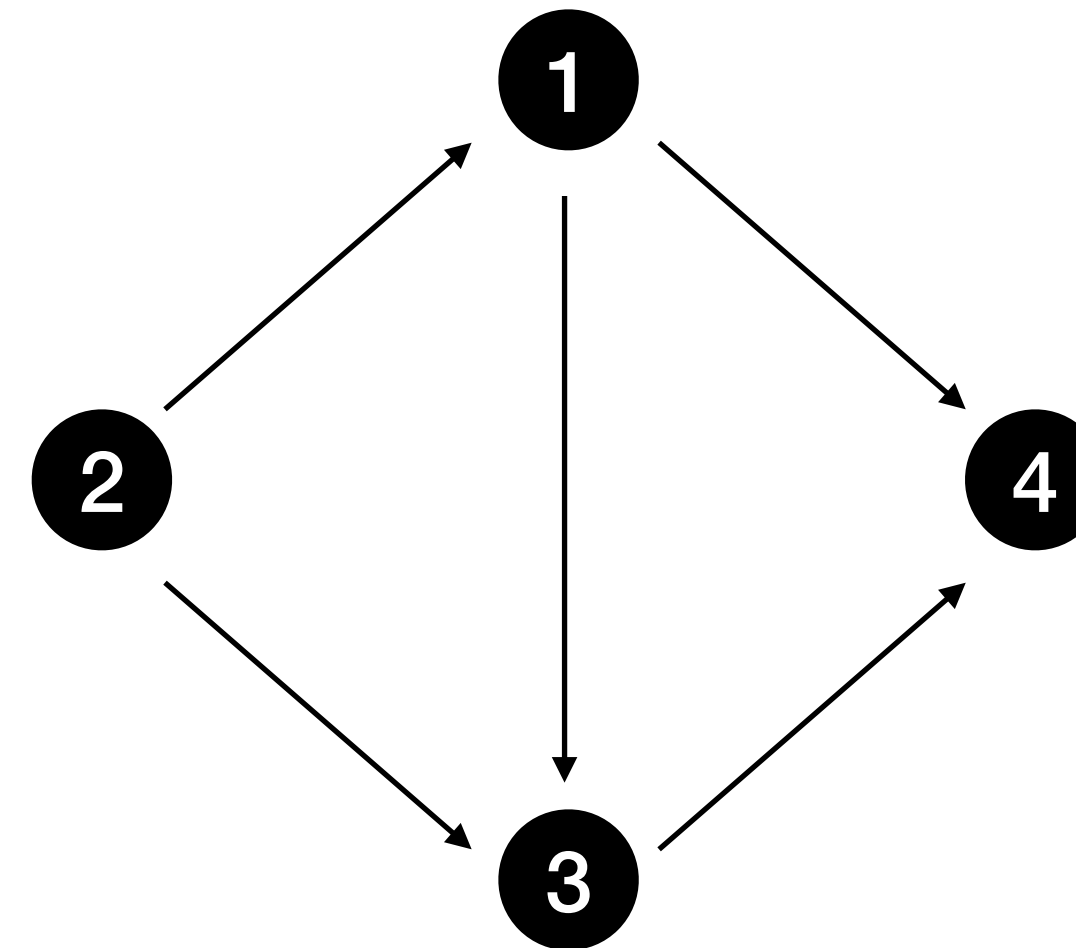
$$\mathcal{N} = \{1, 2, 3, 4\}$$

$$\mathcal{A} = \{(1, 3), (1, 4), (2, 1), (2, 3), (3, 4)\}$$

Directed Graph

Incoming and outgoing arcs

- If $(i, j) \in \mathcal{A}$, we say that arc (i, j) is
 - incident to nodes i and j
 - outgoing from node i
 - Incoming to node j
- $I(i) = \{j \in \mathcal{N} \mid (j, i) \in \mathcal{A}\}$: set of start nodes of arcs that are incoming to node i
- $O(i) = \{j \in \mathcal{N} \mid (i, j) \in \mathcal{A}\}$: set of end nodes of arcs that are outgoing from node i



$$I(4) = \{1, 3\}$$

$$O(3) = \{4\}$$

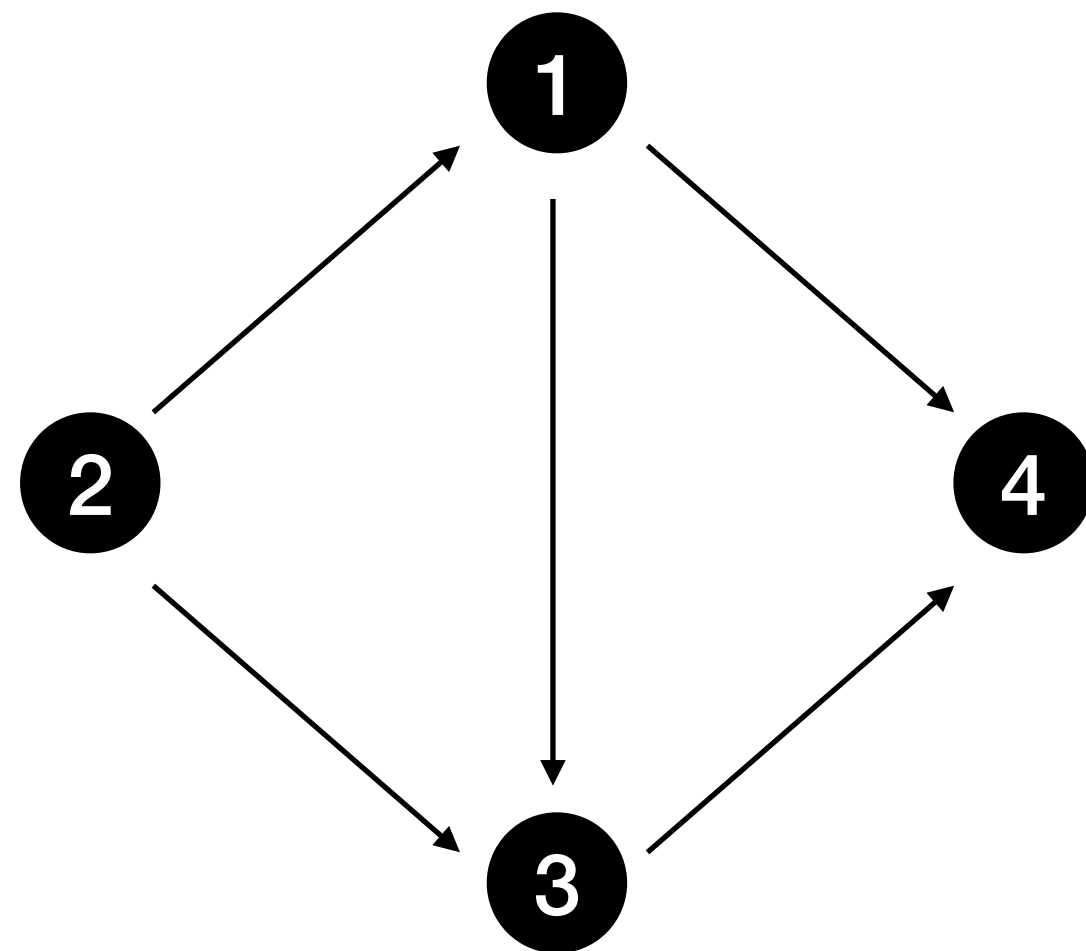
Directed Graph

Node-arc incidence matrix

- A matrix $A = (a_{ik})$ such that

$$a_{ik} = \begin{cases} 1, & \text{if } i \text{ is the start node of the } k\text{-th arc} \\ -1, & \text{if } i \text{ is the end node of the } k\text{-th arc} \\ 0, & \text{otherwise} \end{cases}$$

Arcs: (1,3) (1,4) (2,1) (2,3) (3,4)

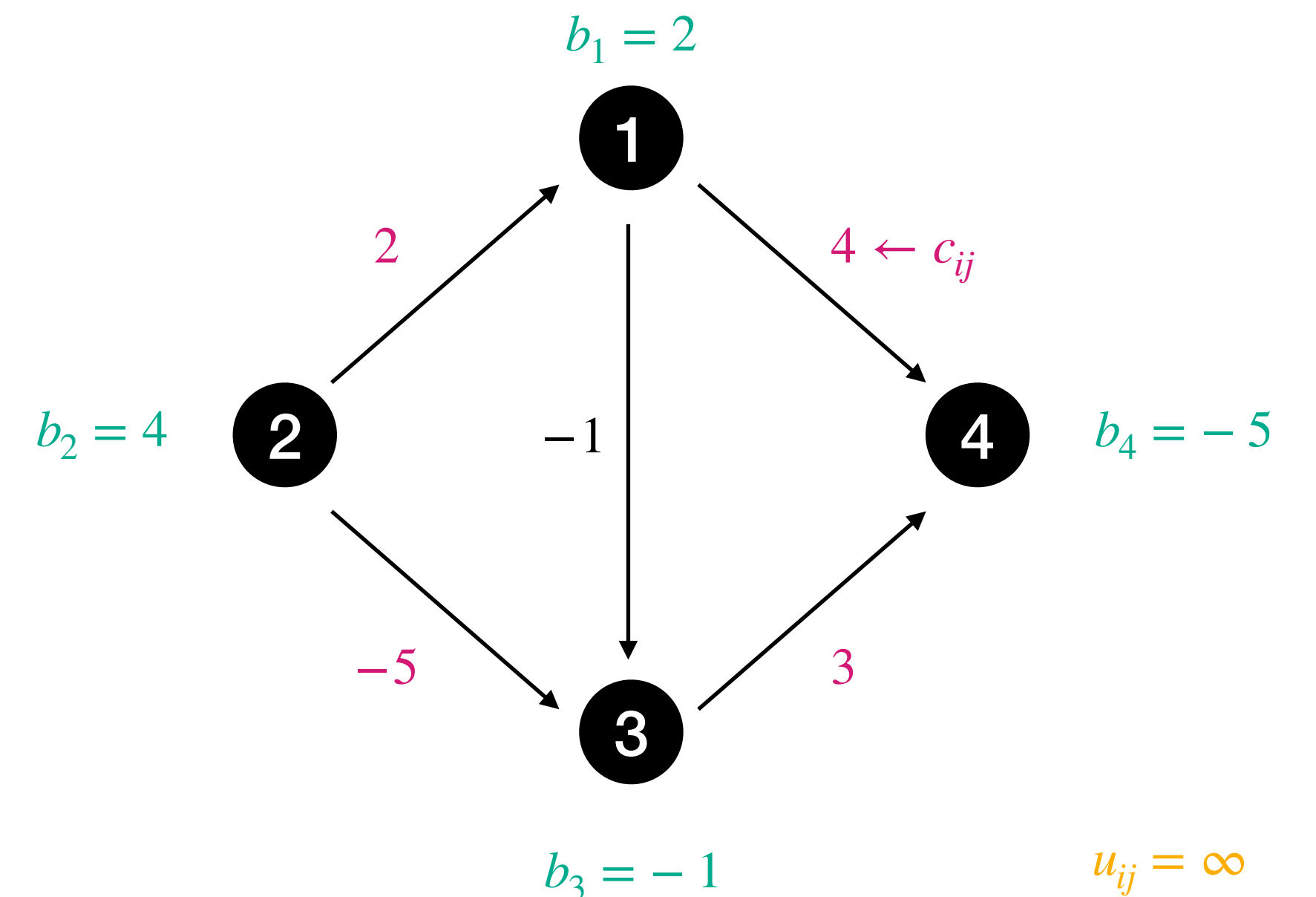


$$A = \begin{pmatrix} a_1^T \\ a_2^T \\ a_3^T \\ a_4^T \end{pmatrix} = \begin{pmatrix} 1 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 0 & -1 \end{pmatrix}$$

Directed Graph Network

A network is a directed graph $G = (\mathcal{N}, \mathcal{A})$ with additional numerical information:

- **supply** in each node i : b_i
 - $b_i > 0$: node i is a source
 - $b_i < 0$: node i is a sink ($|b_i|$ is the demand at node i)
 - assume that $\sum_{i \in \mathcal{N}} b_i = 0$.
- **capacity** of each arc $(i, j) \in \mathcal{A}$: u_{ij}
- **cost** per unit of flow through arc $(i, j) \in \mathcal{A}$: c_{ij}

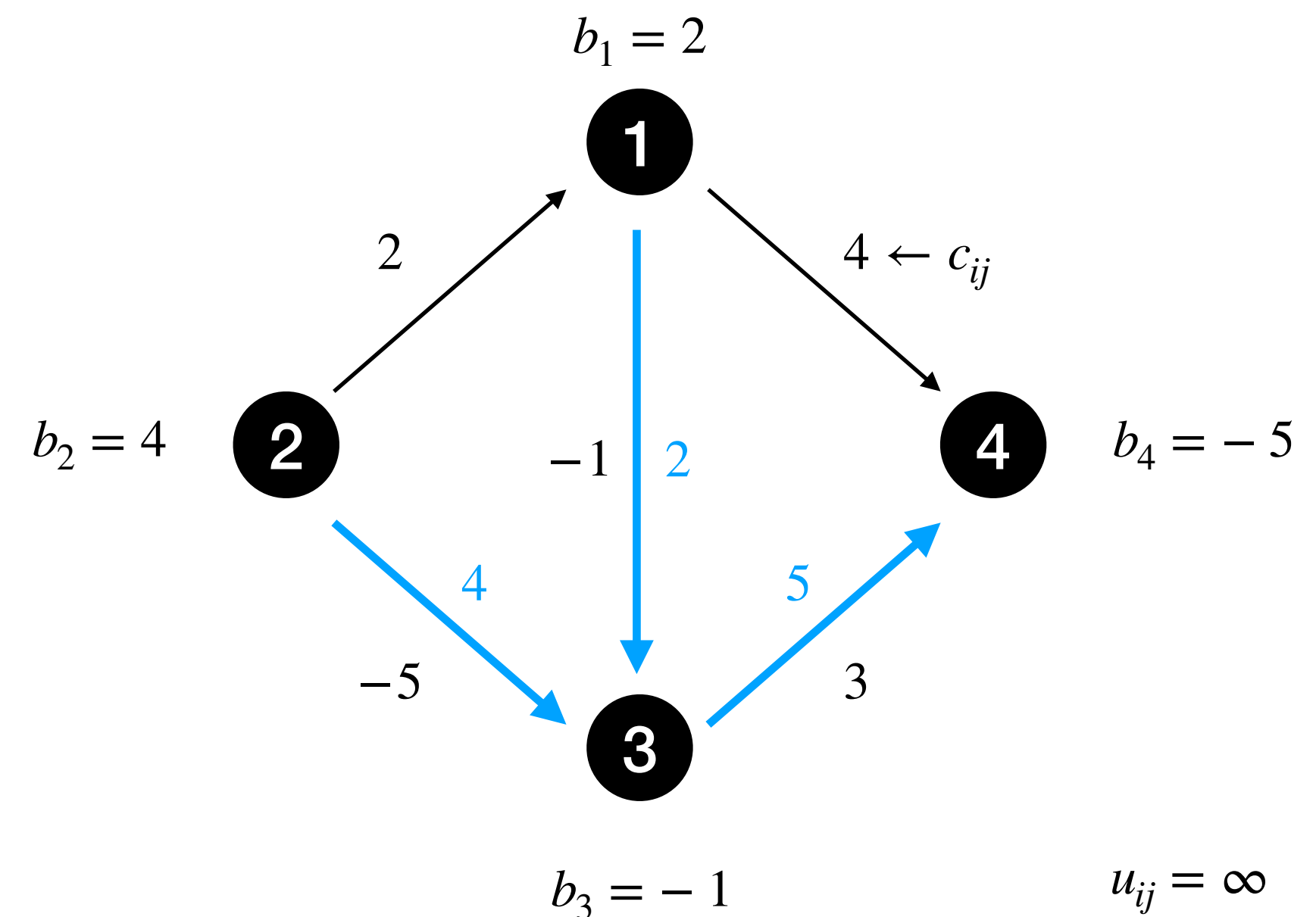
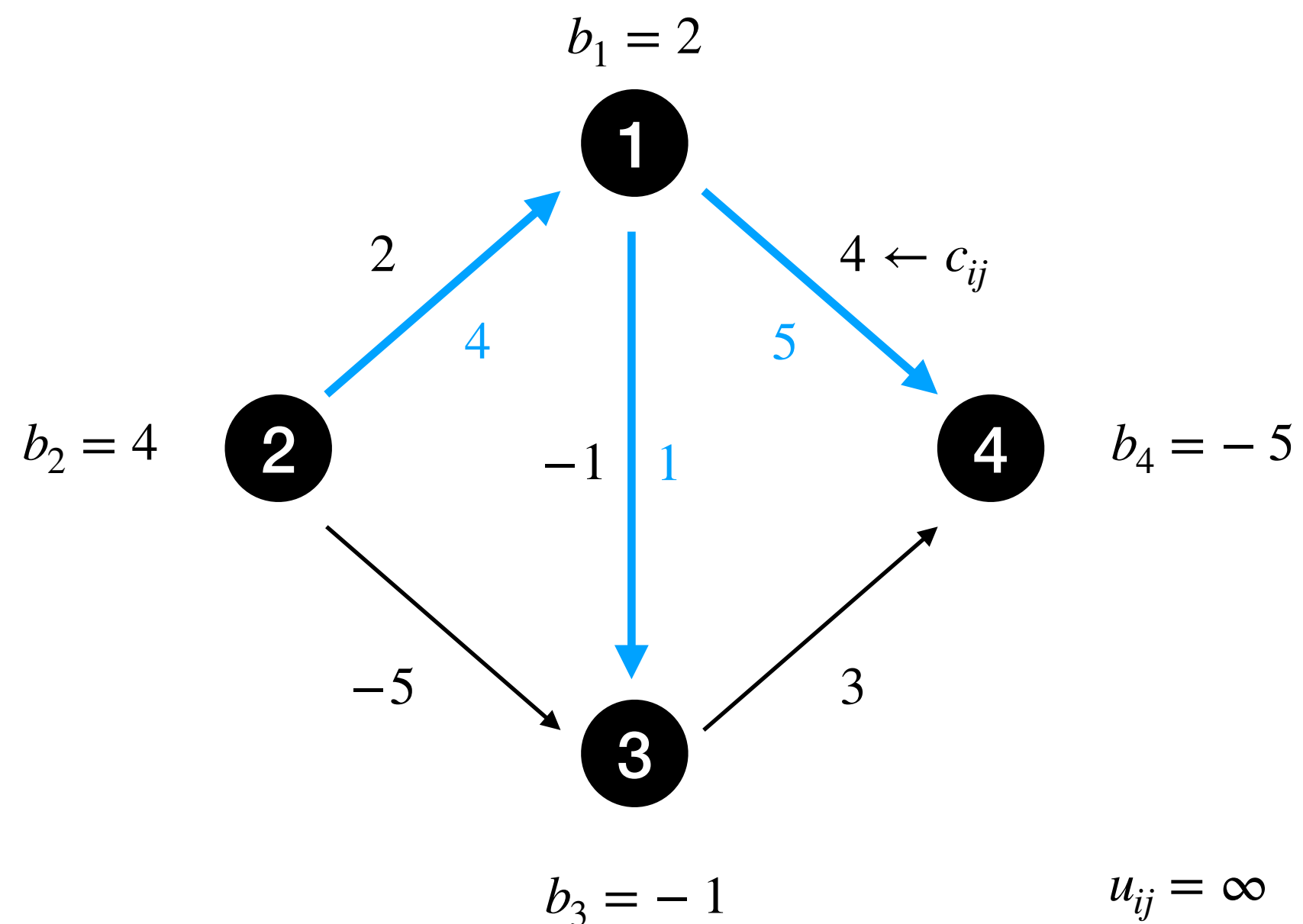


Network Flow Problem

Objective

Compute the amount of **flow** f_{ij} through arc (i,j) for all $(i,j) \in \mathcal{A}$ such that the total cost is minimized.

$$\min \sum_{(i,j) \in \mathcal{A}} c_{ij} f_{ij}$$



Network Flow Problem

Constraints

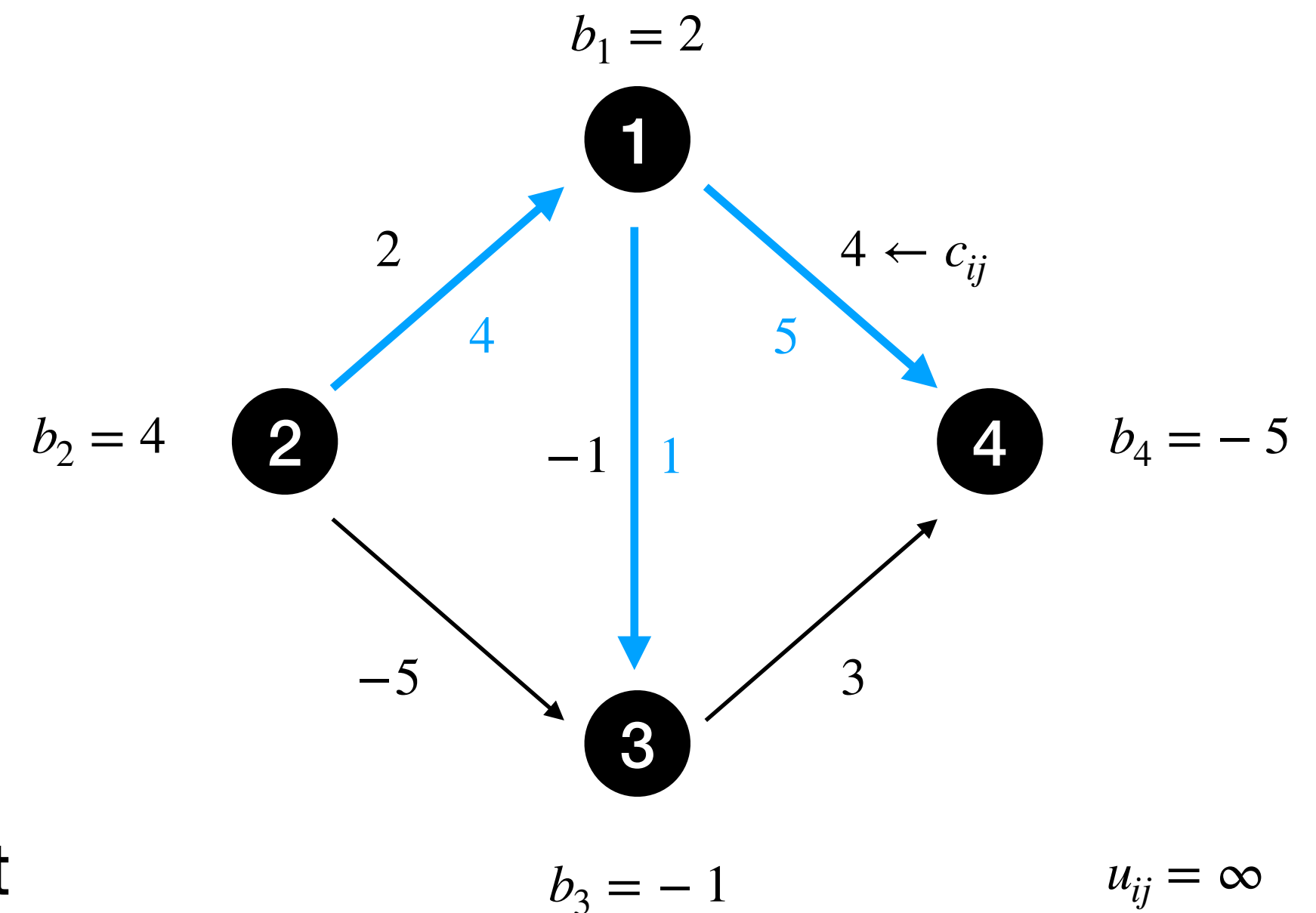
- Flow conservation:

$$b_i + \sum_{j \in I(i)} f_{ji} = \sum_{j \in O(i)} f_{ij}, \quad \forall i \in \mathcal{N}$$

e.g.: $b_1 + f_{21} = f_{13} + f_{14}$, $b_3 + f_{23} + f_{13} = f_{34}$

- Capacity limit: $0 \leq f_{ij} \leq u_{ij}, \quad \forall (i, j) \in \mathcal{A}$

- If a flow satisfies both flow conservation and capacity limit, it is called a **feasible flow**.



Network Flow Problem

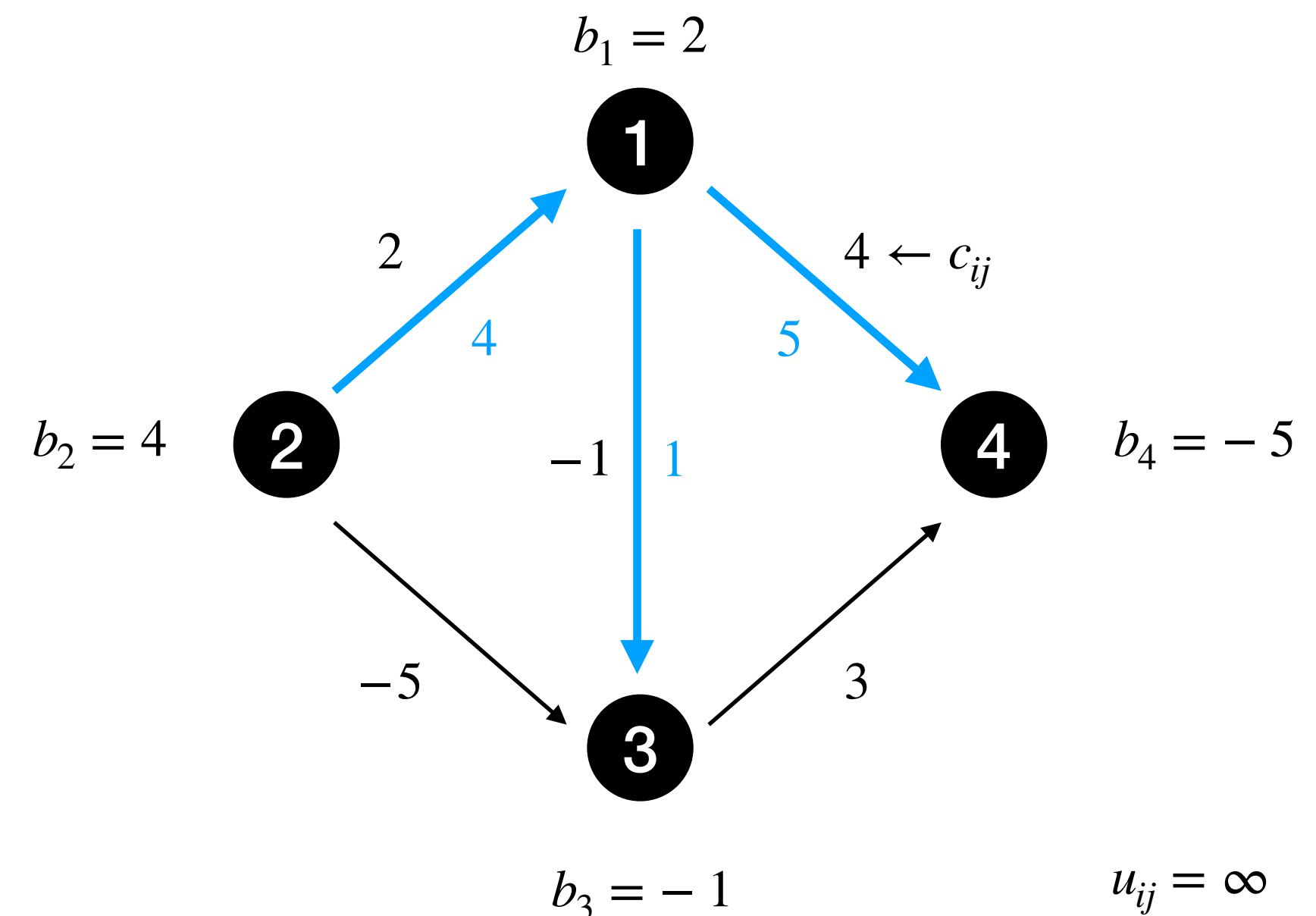
Formulation

Given a network $G = (\mathcal{N}, \mathcal{A})$, node supplies b_i , arc capacities u_{ij} , and arc per unit flow cost c_{ij} , compute a feasible flow f_{ij} such that the total cost is minimized.

$$\min \sum_{(i,j) \in \mathcal{A}} c_{ij} f_{ij}$$

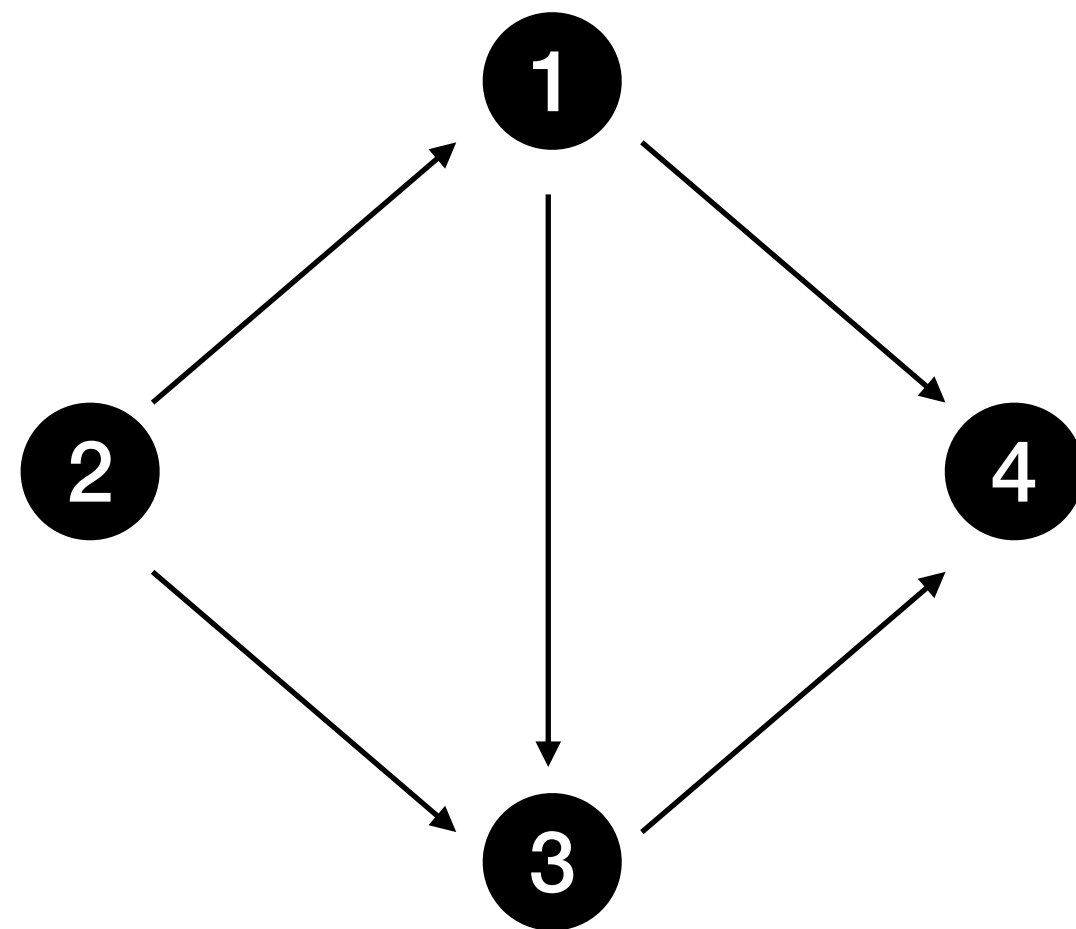
$$\text{S.t.} \quad b_i + \sum_{j \in I(i)} f_{ji} = \sum_{j \in O(i)} f_{ij}, \quad \forall i \in \mathcal{N}$$

$$0 \leq f_{ij} \leq u_{ij}, \quad \forall (i,j) \in \mathcal{A}$$



Network Flow Problem

A concise formulation



Let $f = (f_{13}, f_{14}, f_{21}, f_{23}, f_{34})$ be a flow.

Then, $a_i^T f = ?$

$$\text{E.g.: } a_1^T f = f_{13} + f_{14} - f_{21} = \sum_{j \in O(1)} f_{1j} - \sum_{j \in I(1)} f_{j1}$$

$$A = \begin{pmatrix} a_1^T \\ a_2^T \\ a_3^T \\ a_4^T \end{pmatrix} = \begin{pmatrix} 1 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 0 & -1 \end{pmatrix}$$

Arcs: (1,3) (1,4) (2,1) (2,3) (3,4)

$$a_i^T f = \sum_{j \in O(i)} f_{ij} - \sum_{j \in I(i)} f_{ji}$$

Network Flow Problem

A concise formulation

$$\min \sum_{(i,j) \in \mathcal{A}} c_{ij} f_{ij}$$

$$\text{S.t.} \quad b_i + \sum_{j \in I(i)} f_{ji} = \sum_{j \in O(i)} f_{ij}, \quad \forall i \in \mathcal{N}$$

$$0 \leq f_{ij} \leq u_{ij}, \quad \forall (i,j) \in \mathcal{A}$$

$$a_i^T f = \sum_{j \in O(i)} f_{ij} - \sum_{j \in I(i)} f_{ji} = b_i$$

$$\min \quad c^T f$$

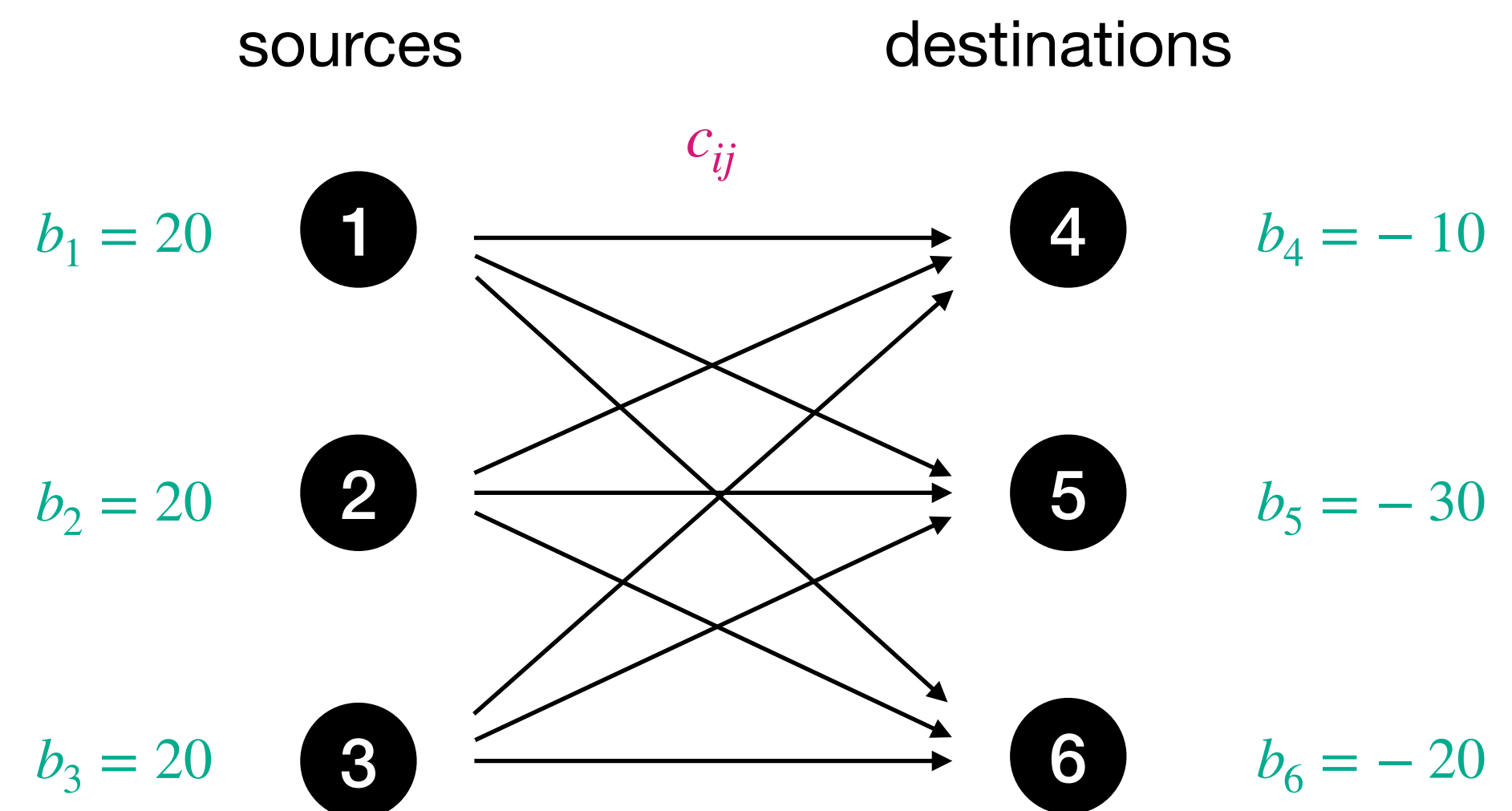
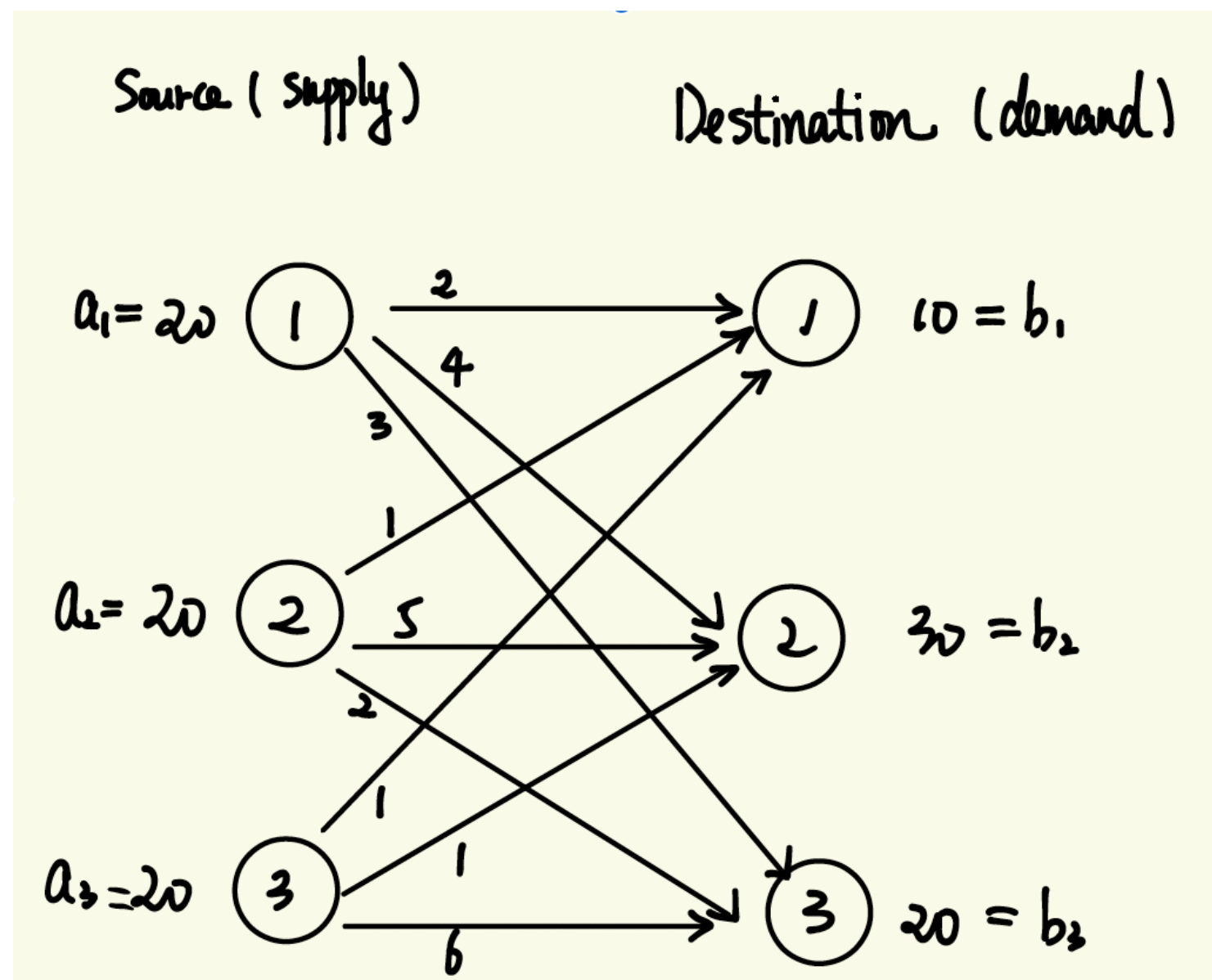
$$\text{S.t.} \quad a_i^T f = b_i, \quad \forall i \in \mathcal{N} \quad \longrightarrow \quad Af = b$$

$$0 \leq f \leq u$$

When $u = \infty$, the problem is a linear program in standard form.

Special Case

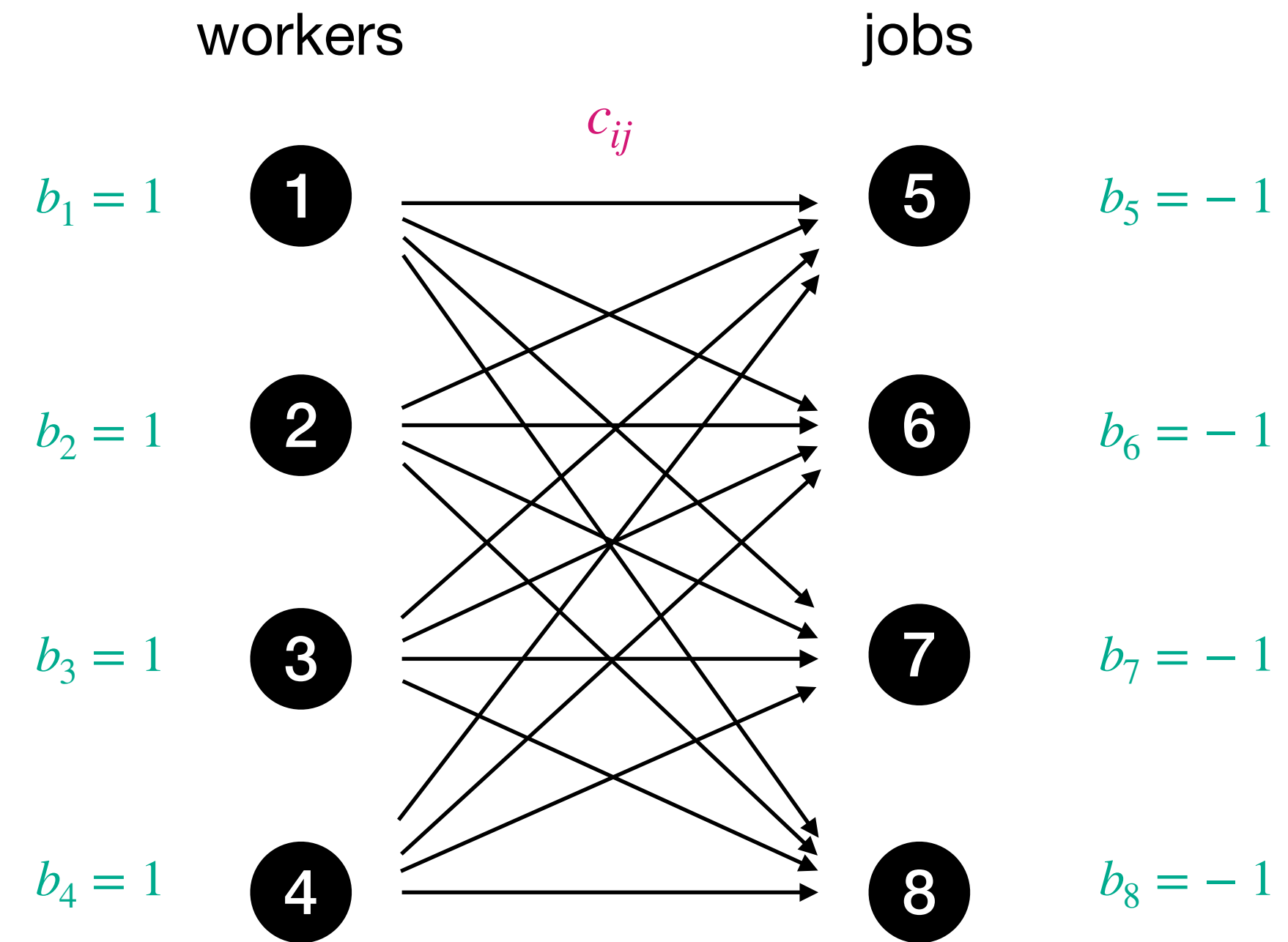
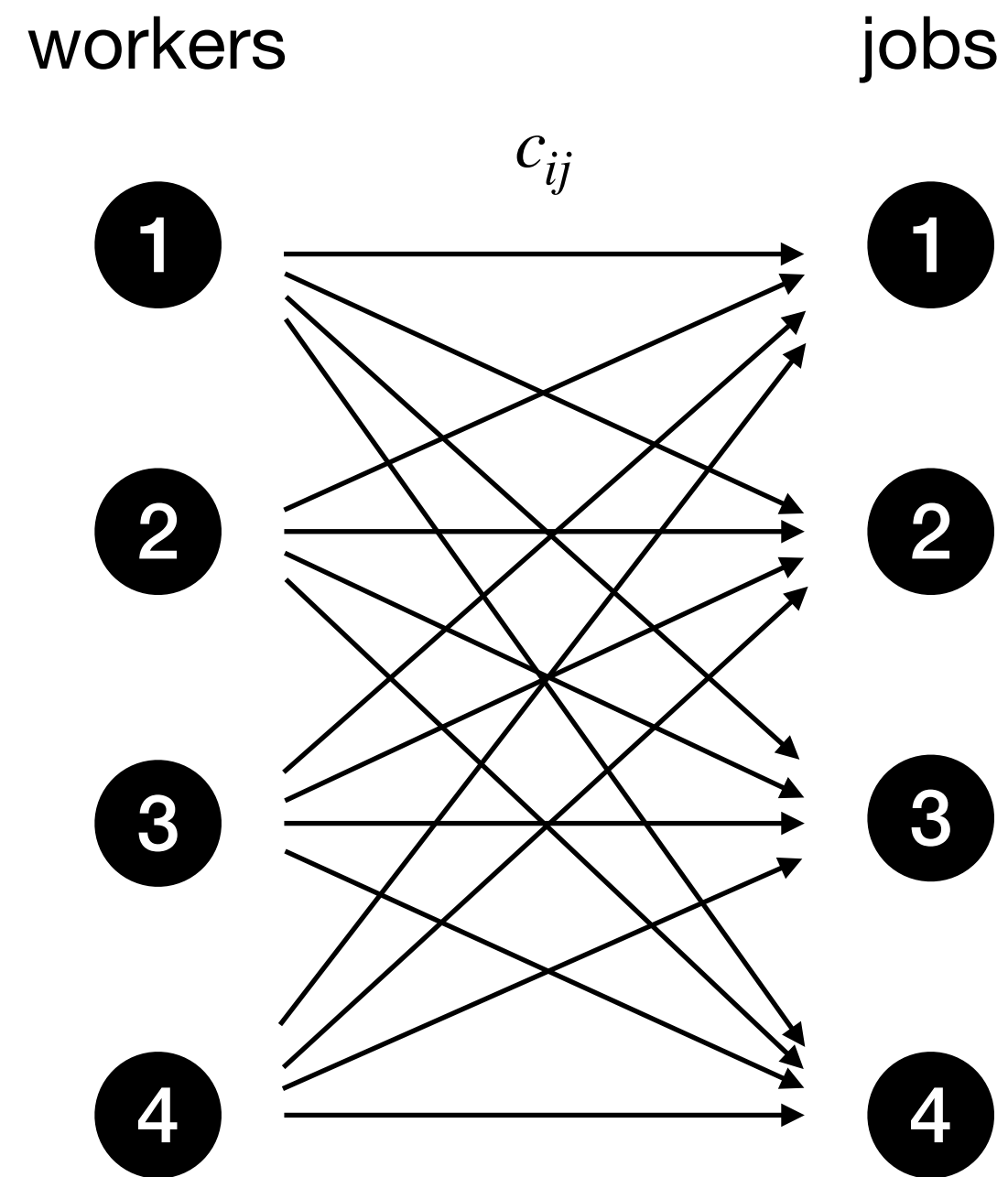
Transportation problem



$$u_{ij} = \infty$$

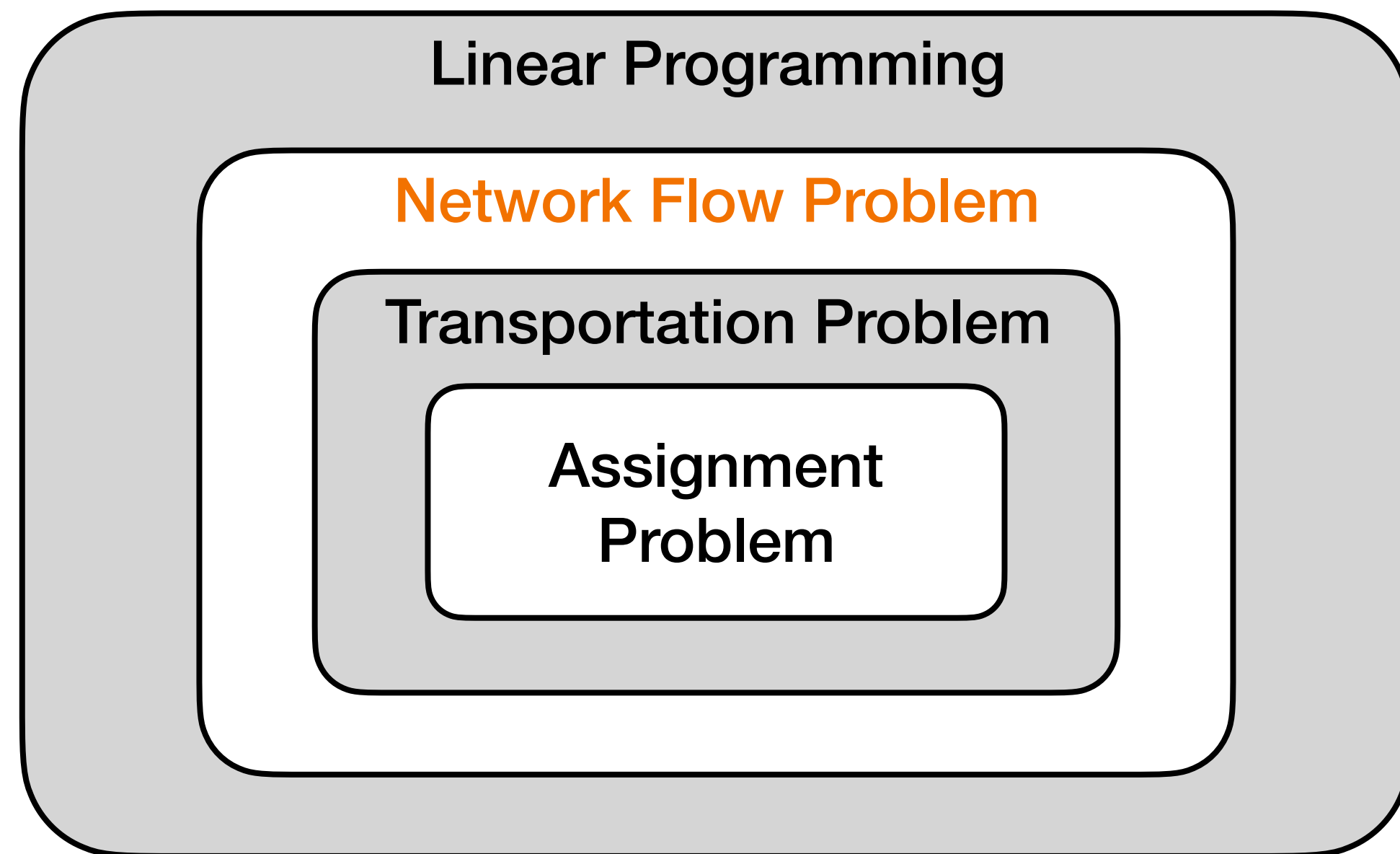
Special Case

Assignment problem



$$u_{ij} = \infty$$

Special Case Relation



- Simplex Method
- Network Simplex Method
- Transportation Algorithm
- Hungarian Algorithm